



Database System Final Project
(Car Showroom Management System)
(BS-SE-F23 Morning)
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(Car Showroom Management System)

(System Introduction)

1. Working

The Car Showroom Management System automates and streamlines the operations of a car showroom. It manages customer information, car inventory, bookings, test drives, sales transactions, warranties, and insurance. By simplifying processes, the system improves business efficiency, reduces manual workload, and enhances customer satisfaction. It also supports decision-making through real-time insights and comprehensive reporting capabilities.

2. Key Features of the System

- Maintains detailed records of customers and their interactions with the showroom.*
- Manages car inventory with attributes like model, price, availability, and specifications.*
- Processes sales transactions, including customer details, payment methods, and delivery schedules.*
- Generates reports based on sales, inventory, and other key metrics to support strategic decision-making.*

3. Description

This system stores data in relational tables with keys and constraints, ensuring consistency. Each table represents a key business object such as a customer, a car, or a sale. Relationships are managed using foreign keys, ensuring accurate record linkage across the database.

4. Purpose

The purpose of this system is to simplify operations, provide accurate record-keeping, and enhance customer satisfaction. It supports business growth by automating tasks and offering insights into performance, inventory, and customer engagement.



(Entities and Their Relations)

1. Entities and Attributes

1. Car

- *Primary key: CarID*
- *Attributes: Make, Model, Year, Colour, EngineType, Transmission, Mileage, Price, VIN, Status*

2. Customer

- *Primary key: CustomerID*
- *Attributes: Name, Phone, Email, Address*

3. Salesperson

- *Primary key: SalespersonID*
- *Attributes: Name, Phone, Email, HireDate, Position, Salary*

4. Sale

- *Primary key: SaleID*
- *Attributes: DateOfSale, FinalPrice, PaymentMethod, CarID (Foreign Key), CustomerID (Foreign Key), SalespersonID (Foreign Key)*

5. TestDrive

- *Primary key: TestDriveID*
- *Attributes: ScheduledDate, Duration, Feedback, CarID (Foreign Key), CustomerID (Foreign Key), SalespersonID (Foreign Key)*

6. ShowRoom

- *Primary key: ShowroomID*

- *Attributes: Name, Location, Phone, AdminID*

7. Inventory

- *Primary key: InventoryID*
- *Attributes: ItemName, Quantity, Category, ShowroomID (Foreign Key)*

8. Staff

- *Primary key: StaffID*
- *Attributes: Name, Role, ShowroomID (Foreign Key), Contact*

9. Payroll

- *Primary key: PayID*
- *Attributes: Salary, PayDate, StaffID (Foreign Key)*

10. Administration

- *Primary key: AdminID*
- *Attributes: Name, Contact, Role, ShowroomID (Foreign Key)*

11. Insurance

- *Primary key: InsuranceID*
- *Attributes: CarID (Foreign Key), CustomerID (Foreign Key), InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount*

12. Booking

- *Primary key: BookingID*
- *Attributes: CustomerID (Foreign Key), CarID (Foreign Key), BookingDate, BookingStatus, AdvanceAmount*

13. Warranty

- *Primary key: WarrantyID*

- *Attributes: CarID (Foreign Key), WarrantyStartDate, WarrantyEndDate, CoverageDetails*

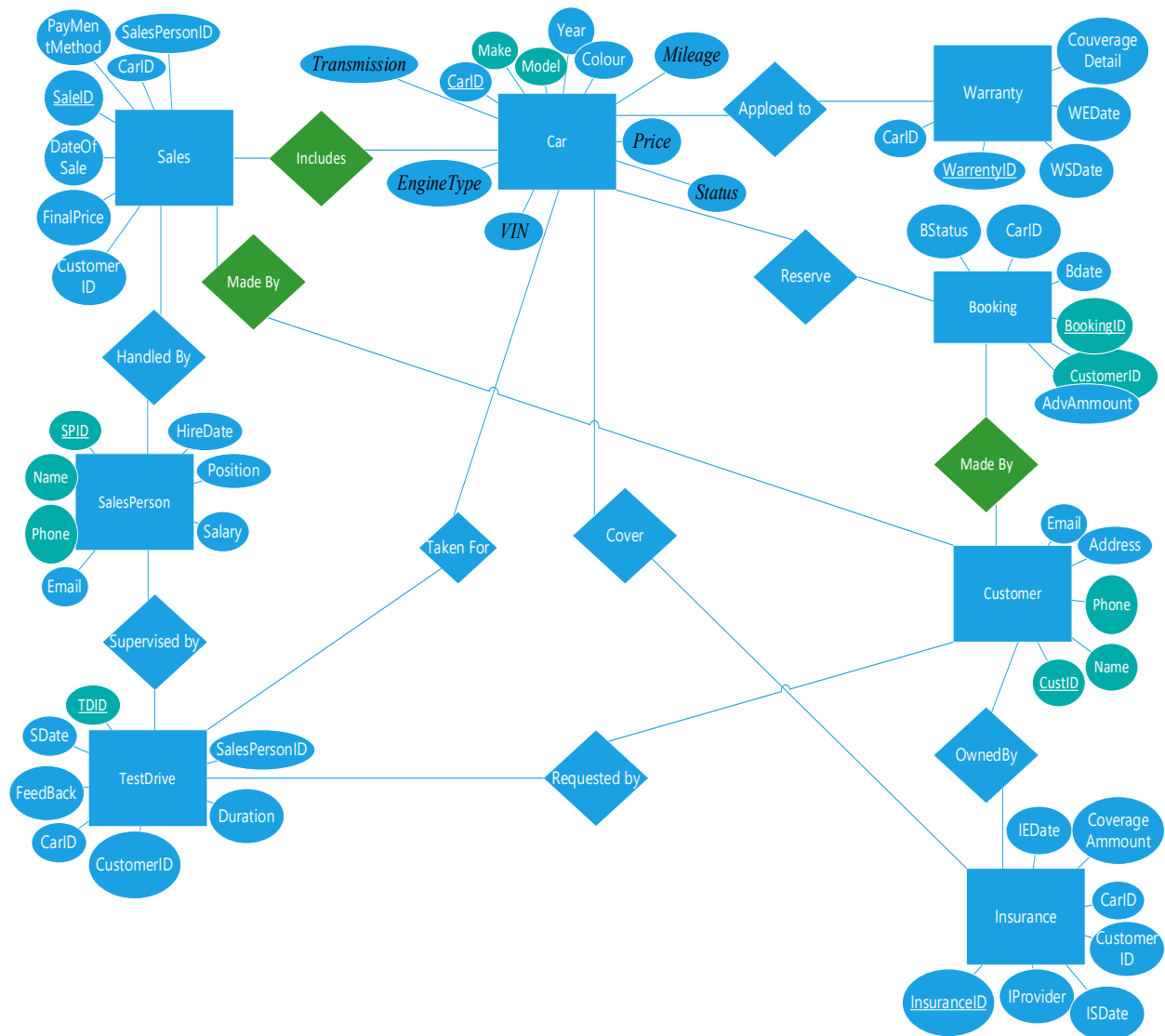
2. Relationships

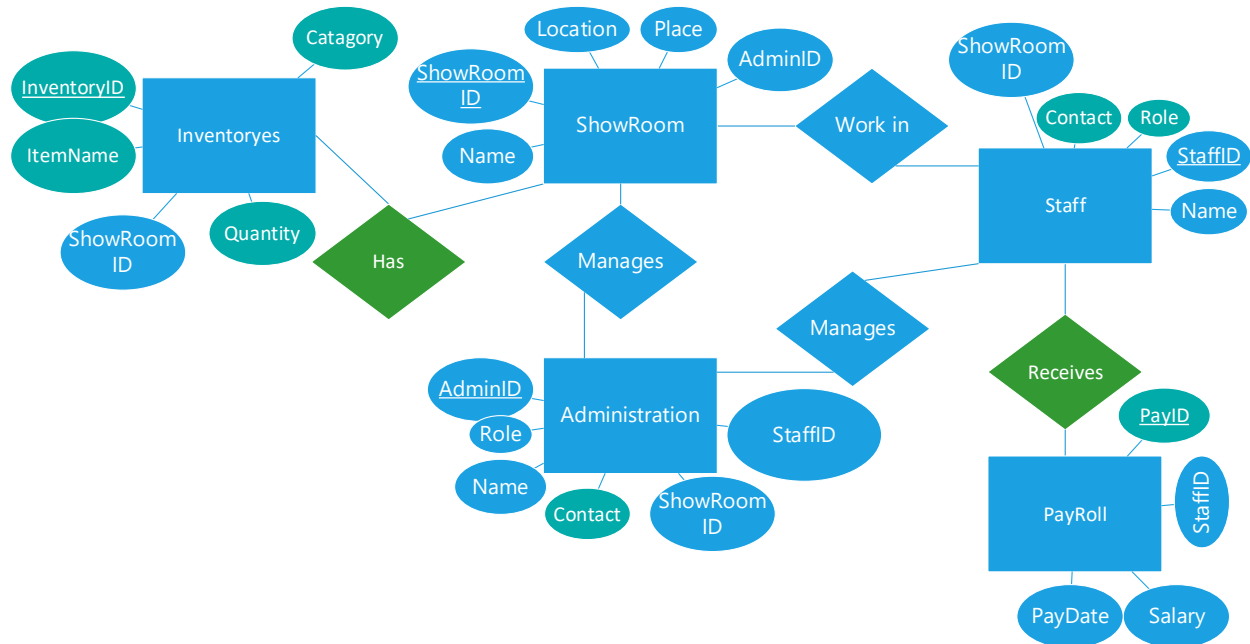
- 1. Sale and Car**
Relationship: Includes
- 2. Sales and Customer**
Relationship: Made by
- 3. Sales and Salesperson**
Relationship: Handled by
- 4. Booking and Customer**
Relationship: Made by
- 5. Booking and Car**
Relationship: Reserves
- 6. TestDrive and Customer**
Relationship: Requested by
- 7. TestDrive and Car**
Relationship: Taken for
- 8. TestDrive and Salesperson**
Relationship: Supervised by
- 9. Warranty and Car**
Relationship: Applies to
- 10. Insurance and Car**
Relationship: Covers
- 11. Insurance and Customer**
Relationship: Owned by
- 12. Payroll and Staff**
Relationship: Receives
- 13. Staff and Showroom**
Relationship: Works in
- 14. Administration and Staff**
Relationship: Manages

15. *Administration and Showroom*
Relationship: *Manages*

16. *Inventory and Showroom*
Relationship: *has*

(Entity Relationship Diagram)





(System Design and Functionalities)

1. Database Design

- *Developed using both top-down and bottom-up design approaches.*
- *Includes an Entity Relationship Diagram (ERD) to visually represent relationships between entities such as Customers, Cars, Sales, and Staff.*
- *Tables are defined with attributes, constraints, and relational keys to ensure consistency and data integrity*

2. Advanced Functionalities

- *Custom Views: Enable simplified access to frequently needed information, such as available cars or customer bookings.*
- *Query-Based Reports: Generate real-time insights including total sales per month, top-performing salespersons, and car availability by showroom.*

(Relational Schema)

1. Top-Down-Approach

1. Car

- (*CarID (PK), Make, Model, Year, Color, EngineType, Transmission, Mileage, Price, VIN, Status*)

2. Customer

- (*CustomerID (PK), Name, Phone, Email, Address*)

3. Salesperson

- (*SalespersonID (PK), Name, Phone, Email, HireDate, Position, Salary*)

4. Sale

- (*SaleID (PK), DateOfSale, FinalPrice, PaymentMethod, CustomerID (FK referencing Customer), CarID (FK referencing Car), SalespersonID (FK referencing Salesperson)*)

5. TestDrive

- (*TestDriveID (PK), ScheduledDate, Duration, Feedback, CarID (FK referencing Car), CustomerID (FK referencing Customer), SalespersonID (FK referencing Salesperson)*)

6. ShowRoom

- (*ShowroomID (PK), Name, Location, Phone, AdminID*)

7. Inventory

- (*InventoryID (PK), ItemName, Quantity, Category, ShowroomID (FK referencing Showroom)*)

8. Staff

- (*StaffID (PK), Name, Role, Contact, ShowroomID (FK referencing Showroom)*)

9. Payroll

- (*PayID (PK), Salary, PayDate, StaffID (FK referencing Staff)*)

10. Administration

- (*AdminID (PK), Name, Contact, Role, ShowroomID (FK referencing Showroom)*)

11. Insurance

- (*InsuranceID (PK), InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount, CarID (FK referencing Car), CustomerID (FK referencing Customer)*)

12. Booking

- (*BookingID (PK), BookingDate, BookingStatus, AdvanceAmount, CustomerID (FK referencing Customer), CarID (FK referencing Car)*)

13. Warranty

- (*WarrantyID (PK), WarrantyStartDate, WarrantyEndDate, CoverageDetails, CarID (FK referencing Car)*)

2. Bottom-Up-Approach

Based on the following criteria, we will do our Bottom Up Approach:

1. A customer may purchase multiple cars (CarID group).
2. A customer may take multiple test drives (TestDriveID group).
3. A customer might have multiple bookings (BookingID group).
4. Showrooms have multiple staff members.
5. Admins could also have multiple contacts/roles? (though that one's fuzzy).
6. Salespersons might be assigned to multiple sales.

1. 0NF

R (CustomerID, Name, Phone, Email, Address, CarID, Make, Model, Year, Color, EngineType, Transmission, Mileage, Price, VIN, Status, SaleID, DateOfSale, FinalPrice, PaymentMethod, SalespersonID, SalespersonName, SalespersonPhone, SalespersonEmail, HireDate, Position, Salary, TestDriveID, ScheduledDate, Duration, Feedback, ShowroomID, ShowroomName, Location, ShowroomPhone, ManagerName, InventoryID, ItemName, Quantity, Category, StaffID, StaffName, StaffRole, StaffContact, PayID, SalaryAmount, PayDate, AdminID, AdminName, AdminContact, AdminRole, InsuranceID, InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount, BookingID, BookingDate, BookingStatus, AdvanceAmount, WarrantyID, WarrantyStartDate, WarrantyEndDate, CoverageDetails)

2. 1NF

1. **Customer**

Customer (CustomerID, Name, Phone, Email, Address)

2. **Car**

Car (CarID, Make, Model, Year, Colour, EngineType, Transmission, Mileage, Price, VIN, Status, WarrantyID, WarrantyStartDate, WarrantyEndDate, CoverageDetails)

3. **Salesperson**

Salesperson (SalespersonID, SalespersonName, SalespersonPhone, SalespersonEmail, HireDate, Position, Salary)

4. **Sale**

Sale (SaleID, CarID, CustomerID, DateOfSale, FinalPrice, PaymentMethod, SalespersonID)

5. **TestDrive**

TestDrive (TestDriveID, CustomerID, CarID, ScheduledDate, Duration, Feedback)

6. **Showroom**

Showroom (ShowroomID, ShowroomName, Location, ShowroomPhone, AdminID)

7. **Inventory**

Inventory (InventoryID, ShowroomID, ItemName, Quantity, Category)

8. **Staff**
Staff (*StaffID, ShowroomID, StaffName, StaffRole, StaffContact, PayID, SalaryAmount, PayDate*)
9. **Administration**
Admin (*AdminID, AdminName, AdminContact, AdminRole*)
10. **Insurance**
Insurance (*InsuranceID, CarID, InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount*)
11. **Booking**
Booking (*BookingID, CustomerID, CarID, BookingDate, BookingStatus, AdvanceAmount*)

3. 2NF

Already in 2NF — there are no partial dependencies in the above 1NF relations.

4. 3NF

- 1) **Customer**
(*CustomerID, Name, Phone, Email, Address*)
- 2) **Car**
(*CarID, Make, Model, Year, Colour, EngineType, Transmission, Mileage, Price, VIN, Status*)
- 3) **Salesperson**
(*SalespersonID, Name, Phone, Email, HireDate, Position, Salary*)
- 4) **Sale**
(*SaleID, DateOfSale, FinalPrice, PaymentMethod, CustomerID, CarID, SalespersonID*)
- 5) **TestDrive**
(*TestDriveID, ScheduledDate, Duration, Feedback, CarID, CustomerID, SalespersonID*)
- 6) **Showroom**
(*ShowroomID, Name, Location, Phone, AdminID*)
- 7) **Inventory**
(*InventoryID, ItemName, Quantity, Category, ShowroomID*)

8) **Staff**

(StaffID, Name, Role, Contact, ShowroomID, AdminID)

9) **Payroll**

(PayID, SalaryAmount, PayDate, StaffID)

10) **Administration**

(AdminID, Name, Contact, Role, ShowroomID)

11) **Insurance**

(InsuranceID, InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount, CarID, CustomerID)

12) **Booking**

(BookingID, BookingDate, BookingStatus, AdvanceAmount, CustomerID, CarID)

13) **Warranty**

(WarrantyID, WarrantyStartDate, WarrantyEndDate, CoverageDetails, CarID)

5. BCNF

Already in BCNF — in all relations, every determinant is a candidate key, so no violations exist.

6. 4NF

Already in 4NF — no relation contains attributes with **multivalued dependencies** not already tied to a candidate key.

7. 5NF

Already in 5NF — no relation needs to be decomposed into **smaller projections** to eliminate anomalies or preserve information.

(Table Description)

1. Customer

Attribute	Data Type	Size	Constraints
CustomerID	Number	-	Primary Key
Name	Varchar2	30	Not Null
Phone	Varchar2	15	Not Null
Email	Varchar2	30	-
Address	Varchar2	50	Not Null

2. Car

Attribute	Data Type	Size	Constraints
CarID	Number	-	Primary Key
Make	Varchar2	20	Not Null
Model	Varchar2	20	Not Null
Year	Number	-	-
Color	Varchar2	10	-
EngineType	Varchar2	15	-
Transmission	Varchar2	10	-
Mileage	Number	-	-
Price	Number	-	-
VIN	Varchar2	17	Unique
Status	Varchar2	10	Check(Status IN ('Available', 'Sold'))

3. SalesPerson

Attribute	Data Type	Size	Constraints
SalespersonID	Number	-	Primary Key
Name	Varchar2	30	Not Null
Phone	Varchar2	15	Not Null
Email	Varchar2	30	-
HireDate	Date	-	Default SYSDATE
Position	Varchar2	20	-
Salary	Number	-	-

4. Sale

Attribute	Data Type	Size	Constraints
SaleID	Number	-	Primary Key
DateOfSale	Date	-	Default SYSDATE
FinalPrice	Number	-	-
PaymentMethod	Varchar2	15	-
CustomerID	Number	-	Foreign Key (CustomerID)
CarID	Number	-	Foreign Key (CarID)
SalespersonID	Number	-	Foreign Key (SalespersonID)

5. TestDrive

Attribute	Data Type	Size	Constraints
TestDriveID	Number	-	Primary Key
ScheduledDate	Date	-	-

Duration	Varchar2	5	-
Feedback	Varchar2	100	-
CarID	Number	-	Foreign Key (CarID)
CustomerID	Number	-	Foreign Key (CustomerID)
SalespersonID	Number	-	Foreign Key (SalespersonID)

6. ShowRoom

Attribute	Data Type	Size	Constraints
ShowroomID	Number	-	Primary Key
Name	Varchar2	30	Not Null
Location	Varchar2	50	Not Null
Phone	Varchar2	15	Not Null
AdminID	Number	-	Foreign Key (AdministrationID)

7. Inventory

Attribute	Data Type	Size	Constraints
InventoryID	Number	-	Primary Key
ItemName	Varchar2	25	Not Null
Quantity	Number	-	Check(Quantity >= 0)
Category	Varchar2	20	Not Null
ShowroomID	Number	-	Foreign Key (ShowroomID)

8. Staff

Attribute	Data Type	Size	Constraints
StaffID	Number	-	Primary Key
Name	Varchar2	30	Not Null
Role	Varchar2	20	Not Null
Contact	Varchar2	15	Not Null
ShowroomID	Number	-	Foreign Key (ShowroomID)

9. Payroll

Attribute	Data Type	Size	Constraints
PayID	Number	-	Primary Key
SalaryAmount	Number	-	-

PayDate	Date	-	Default SYSDATE
StaffID	Number	-	Foreign Key (StaffID)

10. Administration

Attribute	Data Type	Size	Constraints
AdminID	Number	-	Primary Key
Name	Varchar2	30	Not Null
Contact	Varchar2	15	Not Null
Role	Varchar2	20	Not Null
ShowroomID	Number	-	Foreign Key (ShowroomID)

11. Insurance

Attribute	Data Type	Size	Constraints
InsuranceID	Number	-	Primary Key
InsuranceProvider	Varchar2	30	Not Null
InsuranceStartDate	Date	-	-
InsuranceEndDate	Date	-	-
CoverageAmount	Number	-	-
CarID	Number	-	Foreign Key (CarID)
CustomerID	Number	-	Foreign Key (CustomerID)

12. Booking

Attribute	Data Type	Size	Constraints
BookingID	Number	-	Primary Key
BookingDate	Date	-	Default SYSDATE
BookingStatus	Varchar2	20	Check(BookingStatus IN ('Pending', 'Confirmed', 'Cancelled'))
AdvanceAmount	Number	-	-
CustomerID	Number	-	Foreign Key (CustomerID)
CarID	Number	-	Foreign Key (CarID)

13. Warranty

Attribute	Data Type	Size	Constraints
WarrantyID	Number	-	Primary Key

WarrantyStartDate	Date	-	-
WarrantyEndDate	Date	-	-
CoverageDetails	Varchar2	100	-
CarID	Number	-	Foreign Key (CarID)

(SQL Statements)

1. Create Statement
2. Alter Statement
3. Insert Statement
4. Delete Statement
5. Update Statement
6. Select Statement

(Views Creation)

(PL/SQL Statements)

(Tables)